Create Kubernetes Pods using YAML Descriptors

Step 1 - Initialize minikube

- Run *minikube start* and *minikube status* to start and check the stratus of minikube

```
┌(~/Desktop/devops)──────────────────────────────────────────(1rv24mc038_shreerakshahr@asus:pts/1)─┐
└(09:46:25)──> minikube start                                                          ──(Fri,Dec12)─┘
😄  minikube v1.37.0 on Debian bookworm/sid
✨  Using the docker driver based on existing profile
👍  Starting "minikube" primary control-plane node in "minikube" cluster
🚜  Pulling base image v0.0.48 ...
🏃  Updating the running docker "minikube" container ...
🐳  Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔎  Verifying Kubernetes components...
    ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟  Enabled addons: storage-provisioner, default-storageclass
🏄  Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
┌(~/Desktop/devops)──────────────────────────────────────────(1rv24mc038_shreerakshahr@asus:pts/1)─┐
└(09:47:25)──> minikube status                                                         ──(Fri,Dec12)─┘
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
```

Step 2 - Write the ConfigMap file

```
└(09:55:58)──> cat config.yaml                                                         ──(Fri,Dec12)─┘
apiVersion: v1
kind: ConfigMap
metadata:
  name: node-config
data:
  server.js: |
    const http = require("http");
    const port = 3000;

    const server = http.createServer((req,res)=>{
      res.end("Hello from Node server running in Kubernetes");
    });

    server.listen(port, ()=>{
      console.log("Listening on port 3000");
    });
```

Step 3 - Write the Pod YAML file

```
┌─(~/Desktop/devops/lab8)──────────────────────────(1rv24mc038_shreerakshahr@asus:pts/2)─┐
└─(10:47:07)──> cat pod.yaml                                              ──(Fri,Dec12)─┘
apiVersion: v1
kind: Pod
metadata:
  name: node-app
  labels:
    app: node
spec:
  containers:
    - name: node
      image: node:18
      command: ["node", "/usr/src/app/server.js"]
      volumeMounts:
        - name: app-code
          mountPath: /usr/src/app
      ports:
        - containerPort: 3000
  volumes:
    - name: app-code
      configMap:
        name: node-config
```

Step 4 - Write the Node server file

```
┌─(~/Desktop/devops/lab8)──────────────────────────(1rv24mc038_shreerakshahr@asus:pts/1)─┐
└─(09:51:45)──> cat server.js                                             ──(Fri,Dec12)─┘
const http=require("http);
const port = 3000;

const server = http.createServer((req,res)=>{
        res.end("Hello from Node app hosted on Kubernetes");
});

server.listen(port, ()=>console.log("Listening on port 3000"));
```

Step 5 - Apply the written YAML files

```
┌─(~/Desktop/devops/lab8)──────────────────────────(1rv24mc038_shreerakshahr@asus:pts/1)─┐
└─(10:00:57)──> kubectl apply -f config.yaml                              ──(Fri,Dec12)─┘
configmap/node-config created
```

```
┌─(~/Desktop/devops/lab8)──────────────────────────(1rv24mc038_shreerakshahr@asus:pts/1)─┐
└─(10:01:38)──> kubectl apply -f pod.yaml                                 ──(Fri,Dec12)─┘

pod/node-app created
```

```
┌─(~/Desktop/devops/lab8)──────────────────────────(1rv24mc038_shreerakshahr@asus:pts/1)─┐
└─(10:02:02)──> kubectl get pods                                          ──(Fri,Dec12)─┘
NAME                               READY   STATUS    RESTARTS      AGE
nginx-deployment-6f9664446b-z7ld2  1/1     Running   1 (16m ago)   18m
node-app                           1/1     Running   0             113s
```

Step 6 - Expose port 3000 for the server

```
┌(~/Desktop/devops/lab8)──────────────────────────────(1rv24mc038_shreerakshahr@asus:pts/1)─┐
└(10:03:33)──> kubectl port-forward pod/node-app 3000:3000                     ──(Fri,Dec12)─┘
Forwarding from 127.0.0.1:3000 -> 3000
Forwarding from [::1]:3000 -> 3000
```

Output -